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Assignment: Assignment 01

Section: B

Course: Cloud Computing

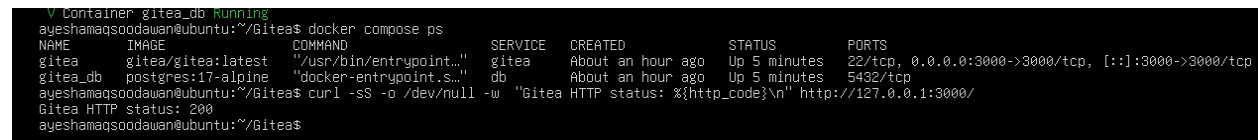
Topic: Git, Gitea, GitHub, Git LFS, and GitHub Pages

1. Task 1 – Gitea

1.1 Gitea Installation

- Used Ubuntu Server as the working environment.
- Installed Docker and Docker Compose.
- Started the Gitea and PostgreSQL containers.
- Verified that both services were running successfully.
- Verified Gitea using the local HTTP request.
- Accessed Gitea through the Ubuntu server IP.
- Configured PostgreSQL as the database.
- Completed the Gitea installation and created the administrator account.

Figure 1 – Gitea running on Ubuntu Server



```
V Container gitea_db running
ayeshamagsoodawane@ubuntu:~/Gitea$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED          STATUS          PORTS
gitea                gitea/gitea:latest  "/usr/bin/entrypoint..." gitea      About an hour ago Up 5 minutes    22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db             postgres:17-alpine  "docker-entrypoint.s..." db         About an hour ago Up 5 minutes    5432/tcp
ayeshamagsoodawane@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
ayeshamagsoodawane@ubuntu:~/Gitea$
```

task1_gitea_running.png

1.2 Push Repository to Gitea

- Created a public Gitea repository named Assignment01.
- Created a local Git repository on Ubuntu.
- Initialized the repository with the main branch.
- Created README.md containing:
 - o Full Name
 - o Registration Number
- Committed the README file.
- Added Gitea as a remote named gitea.
- Successfully pushed the main branch to Gitea.

Figure 2 – Successful push to Gitea

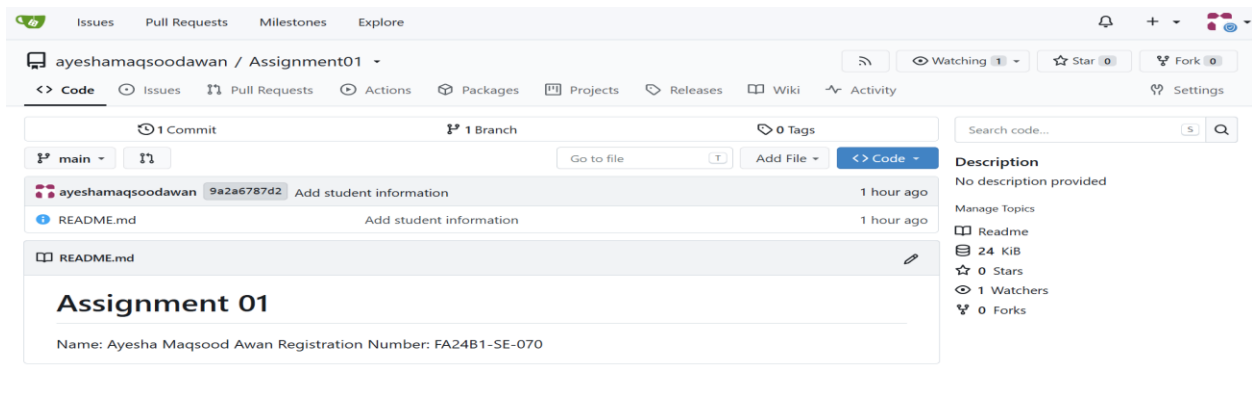
```
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git remote -v
gitea  http://ayeshamaqsoodawan@127.0.0.1:3000/ayeshamaqsoodawan/Assignment01.git (fetch)
gitea  http://ayeshamaqsoodawan@127.0.0.1:3000/ayeshamaqsoodawan/Assignment01.git (push)
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ayeshamaqsoodawan@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 307.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/ayeshamaqsoodawan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
ayeshamaqsoodawan@ubuntu:~/Assignment01$ _
```

task1_gitea_push.png

1.3 Gitea Repository Verification

- Opened the Assignment01 repository in Gitea.
- Verified:
 - o Repository name
 - o Owner
 - o Public visibility
 - o README file
 - o Student information

Figure 3 – Gitea Assignment01 repository



task1_gitea_repository.png

Result

- Gitea was successfully installed.
- Repository was successfully created and pushed.

2. Task 2 – GitHub

2.1 Configure Git Remotes

- Created a public GitHub repository named Assignment01.
- Used the same local repository from Task 1.
- Added GitHub as a second remote named github.
- Verified both remotes using git remote -v.
- The repository now had:
 - o gitea remote
 - o github remote

Figure 4 – Gitea and GitHub remotes

```
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git remote -v
gitea    http://ayeshamaqsoodawan@127.0.0.1:3000/ayeshamaqsoodawan/Assignment01.git (fetch)
gitea    http://ayeshamaqsoodawan@127.0.0.1:3000/ayeshamaqsoodawan/Assignment01.git (push)
github   https://github.com/ayeshamaqsoodawan/Assignment01.git (fetch)
github   https://github.com/ayeshamaqsoodawan/Assignment01.git (push)
ayeshamaqsoodawan@ubuntu:~/Assignment01$ _
```

2.2 Push to GitHub

- Pushed the main branch to GitHub.
- GitHub authentication was completed successfully.
- Push completed without errors.

Figure 5 – Successful GitHub pushtask2_remotes.png

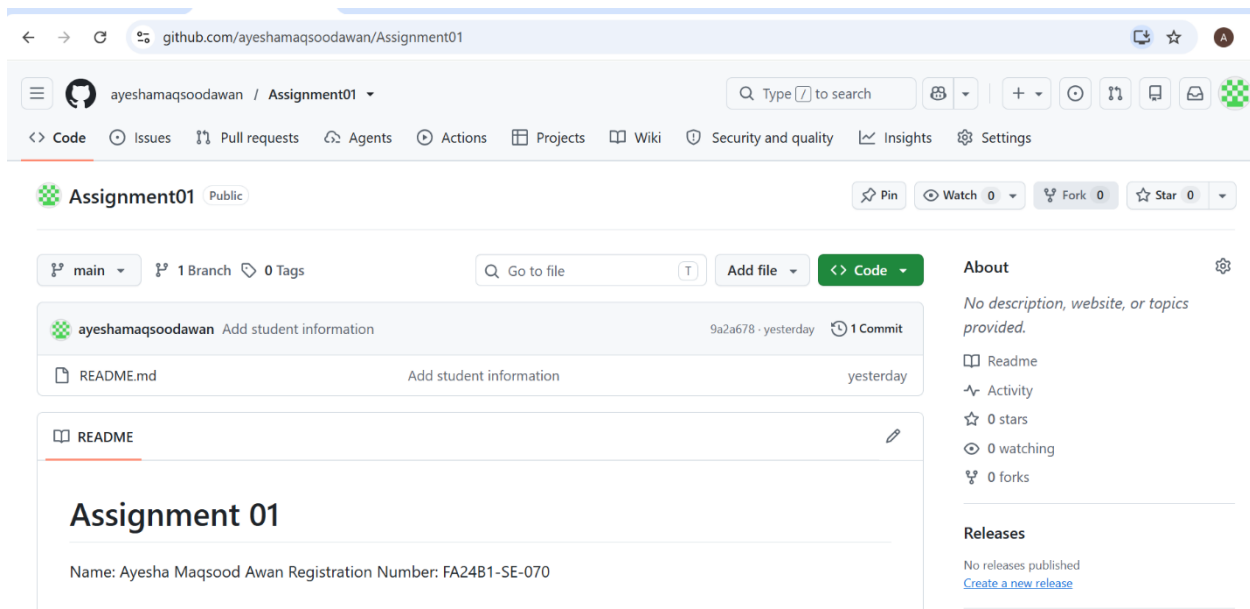
```
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': ayeshamaqsoodawan
Password for 'https://ayeshamaqsoodawan@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 38.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayeshamaqsoodawan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
ayeshamaqsoodawan@ubuntu:~/Assignment01$ _
```

task2_github_push.png

2.3 GitHub Repository Verification

- Opened the GitHub Assignment01 repository.
- Verified:
 - o Repository name
 - o Owner
 - o Public visibility
 - o README file
 - o Student information

Figure 6 – GitHub Assignment01 repository



task2_github_repository.png

Result

- The same local repository was successfully pushed to both Gitea and GitHub.

3. Task 3 – Git LFS

3.1 Git LFS Setup

- Initialized Git LFS in the Assignment01 repository.
- Verified the installed Git LFS version.
- Configured .bin files to be tracked using Git LFS.
- Verified the .gitattributes file.

Figure 7 – Git LFS setup

```
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
ayeshamaqsoodawan@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
ayeshamaqsoodawan@ubuntu:~/Assignment01$ _
```

task3_lfs_setup.png

3.2 Large File Creation

- Created three binary files:
 - o large-file-1.bin – 101 MiB
 - o large-file-2.bin – 102 MiB
 - o large-file-3.bin – 103 MiB
- Added the files to Git.
- Verified that all three files were tracked by Git LFS.

Figure 8 – Git LFS tracked files

```
*.bin filter=lfs diff=lfs merge=lfs -text
ayeshamaqsoodawan@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
ayeshamaqsoodawan@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
ayeshamaqsoodawan@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
ayeshamaqsoodawan@ubuntu:~/Assignment01$ _
```

task3_lfs_files.png

3.3 Git LFS Push

- Committed the three large files.
- Pushed the commit to GitHub.
- Git LFS uploaded the large file objects successfully.
- Push completed successfully.

Figure 9 – Successful Git LFS push

```
ayeshamaqsoodawan@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': ayeshamaqsoodawan
Password for 'https://ayeshamaqsoodawan@github.com':
Username for 'https://github.com':
Password for 'https://github.com':
Username for 'https://github.com': ayeshamaqsoodawan
Password for 'https://ayeshamaqsoodawan@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 1.7 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 802 bytes | 267.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayeshamaqsoodawan/Assignment01.git
 9a2a678..ad54d41  main -> main
ayeshamaqsoodawan@ubuntu:~/Assignment01$
```

task3_lfs_push.png

Result

- Three files larger than 100 MiB were successfully managed and pushed using Git LFS.

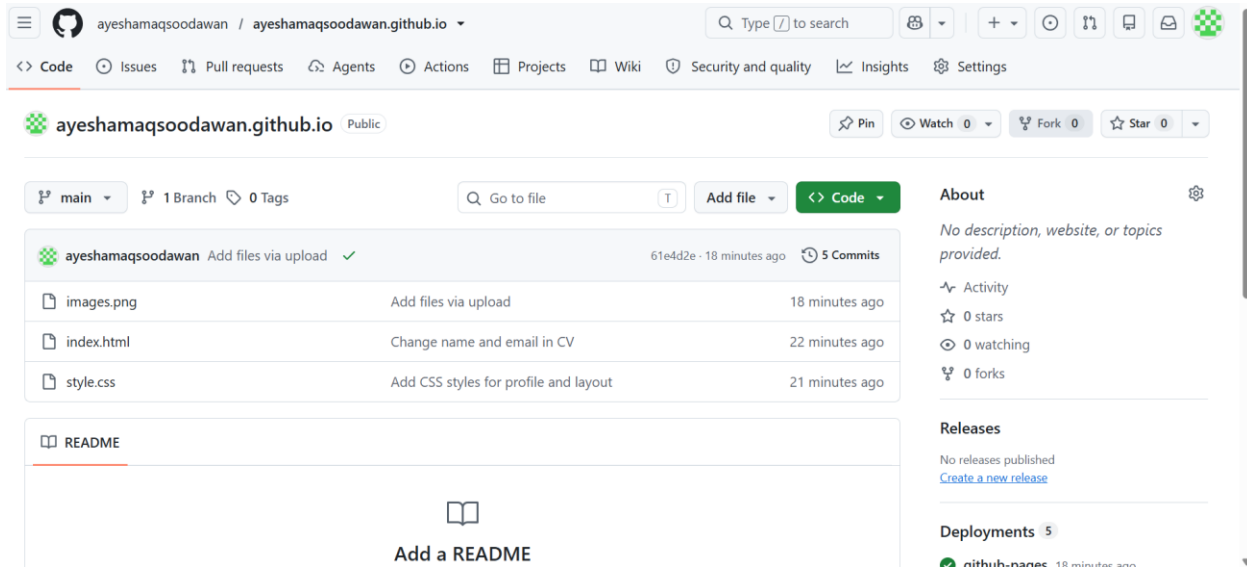
4. Task 4 – GitHub Pages Portfolio

4.1 Portfolio Creation

- Created a personal portfolio website using HTML and CSS.
- Created the following files:
 - o index.html
 - o styles.css
- Added the following sections:
 - o About Me
 - o Education
 - o Skills
 - o Projects
- Used a green, white, and black design.

- Created a GitHub repository named salehataj.github.io.
- Pushed the portfolio to the main branch.

Figure 10 – Portfolio GitHub repository

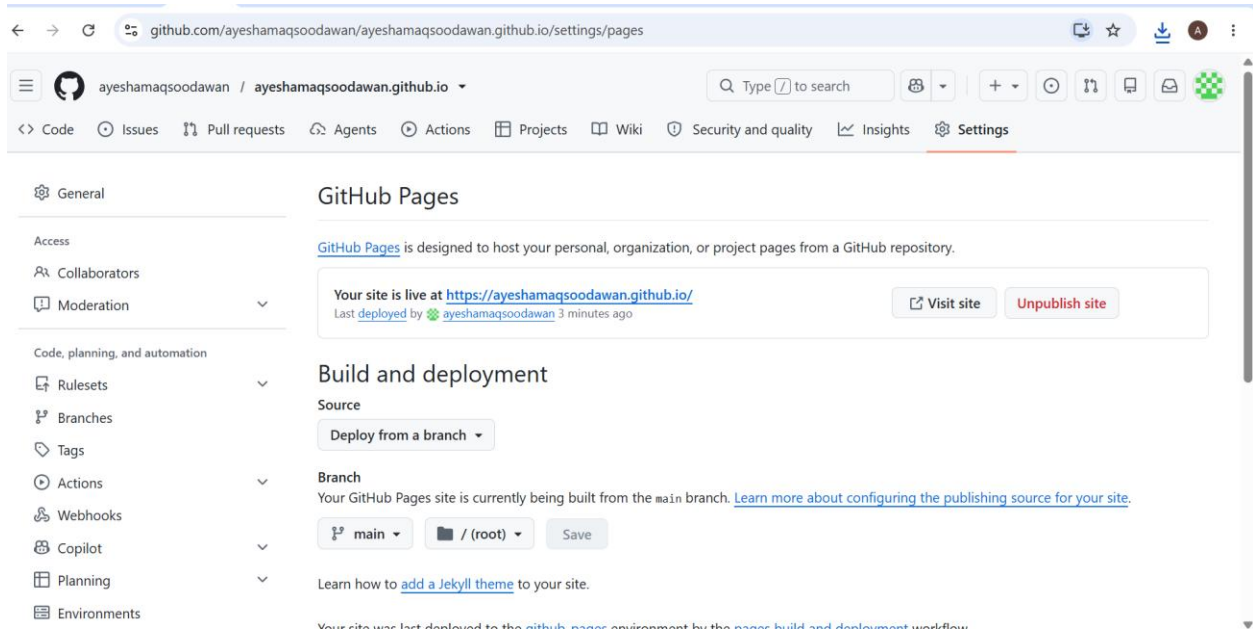


4.2 GitHub Pages Deployment

- Opened the repository's Settings → Pages.
- Selected deployment from the main branch.
- Selected the root directory / (root).
- Saved the configuration.
- GitHub Pages successfully deployed the website.
- Generated live website URL:

<https://salehataj.github.io/>

Figure 11 – GitHub Pages deployment



task4_pages_repository.png

4.3 Live Portfolio

- Opened the GitHub Pages URL in the browser.
- Verified that the website was publicly accessible.
- Confirmed that the portfolio content was displayed correctly.

Figure 12 – Live portfolio website



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Nationality: Pakistani

Date of birth: 14-07-2001

Gender: Female

Marital Status: Single

Objective

To obtain a responsible position in a well-established organization where I can contribute to the organization's success through my good work habits, flexibility, and commitment to excellence. I am a competent and willing person with the ability to learn quickly, work well under pressure, and be effective in team-based work while always striving for personal and professional growth.

Experience

07/2024 - 09/2024

Video Editor
Internship with Global Education Foundation

Result

- The portfolio website was successfully created and published using GitHub Pages.